

Leiomyosarcoma (LMS) & Uterine LMS Patient & Caregiver Guide



What LMS Is

- **Leiomyosarcoma (LMS)** = rare, aggressive cancer of **smooth muscle cells**
- **Uterine LMS (uLMS)**
 - Starts in uterine muscle (myometrium)
 - Often **more aggressive** than other uterine cancers
 - Frequently spreads through blood → **lungs common**

Key Biology (Why It's Hard to Treat)

- Not one disease → **highly complex genomics**
- **No single dominant mutation**
- Common features:
 - Genomic instability
 - Chromosomal gains/losses
 - Tumor suppressor loss

Key pathways:

- *TP53* → cell cycle control
- *RB1* → tumor suppression
- *ATR* → chromatin regulation
- DNA repair defects (subset)

Clinical Patterns

- **Uterine LMS**
 - More aggressive
 - Hormone receptors (ER/PR) may matter
- **Extra-uterine LMS**
 - Location-dependent behavior (abdomen, limbs)
- **Emerging molecular patterns**
 - TP53 + RB1 loss (common)
 - DNA repair-deficient (rare, but important)

Why NGS Testing Matters

Even without obvious “targets,” NGS helps by:

- Finding **rare actionable mutations**
- Identifying **clinical trial options**
- Detecting **DNA repair defects**
- Guiding future decisions as disease evolves

How to Read an NGS Report

1. **Somatic mutations**
 - Tumor-only changes (e.g., TP53, RB1)
 - Usually **not directly targetable**
2. **TMB (Tumor Mutational Burden)**
 - Usually low
 - High (rare) → may help immunotherapy
3. **MSI status**
 - Usually stable (MSS)
 - MSI-high (rare) → important for treatment
4. **Copy number changes**
 - Very common → reflects instability
5. **Gene fusions**
 - Rare in LMS
 - If present → may change treatment options
6. **VUS (Variants of Uncertain Significance)**
 - Unknown meaning → **do not act on alone**

Treatment Overview

First-line:

- Doxorubicin-based therapy
- ± Gemcitabine / Docetaxel

Later lines:

- Pazopanib
- Trabectedin
- Eribulin

Example: STC-15 (METTL3 Inhibitor)

- Targets RNA methylation (epitranscriptomics)
- Blocks METTL3 → affects:
 - Cancer stem cells
 - Tumor differentiation
- Goal: reduce tumor “self-renewal”
- Potential relevance in aggressive LMS biology

Clinical Trials (Strongly Recommended)

LMS care often includes **early trial consideration**

Common approaches:

- DNA damage response drugs
- Cell cycle inhibitors
- Immunotherapy combinations
- Epigenetic therapies

What Matters Most in Real Care

- Treatment at a **sarcoma specialty center**
- **Early NGS testing**
- **Re-biopsy** if disease changes
- Stay **clinical trial ready**
- Track **symptoms + quality of life**

Trusted Resources

- [National Leiomyosarcoma Foundation \(NLMSF\)](https://www.nlmsf.org)



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Sarcoma & LMS Pathways

Care Guide - Quick Tips for Patients & Families

Trusted guidance for navigating diagnosis, treatment, and leiomyosarcoma

[www.NLMSF.org](https://www.nlmsf.org)